

# HIMANSHU RATHOD

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## EDUCATION

DREXEL UNIVERSITY , *Philadelphia, PA*

March 2023

Master of Science in Computer Science

GPA: 3.55/4.0

Selected Coursework: Machine Learning, Computer Vision, Artificial Intelligence, Software Design, Human Computer Interaction, Database Management Systems, Data Analysis and Interpretation, Data Acquisition and Pre-Processing , Data Structures and Algorithms, System Basics, Design of Graphical User Interfaces, Operating Systems

RAMAIAH INSTITUTE OF TECHNOLOGY (MSRIT), *Bangalore, India*

June 2019

Bachelor of Engineering in Mechanical Engineering

GPA: 7.75/10

Selected Coursework: Artificial Intelligence, Python Programming

## SKILLS

Programming Languages: Python, C++ ,C, C#, Java, SQL, Unix Scripting (bash), Node.js, MongoDB

Frameworks: Azure Function Apps, Azure Logic Apps, Azure Service Bus, Azure APIM, Azure Blob storage,

Azure Key vault, Azure Management Portal , .NET MVC, .NET Core, Data Grip, Azure CLI, PyTorch,

TensorFlow, Keras, Pandas, Scikit-learn, NumPy, SciPy, Matplotlib, NextJS, ReactJs, Rasa, Express,

Convolutional Neural Networks, Recurrent Neural Networks, Auto Encoders

Generative adversarial networks, Spring Boot, Tableau , HTML, CSS, Bootstrap, Natural Language Processing

Version Control Systems: Git, Perforce, Bitbucket

## EXPERIENCE

OLD REPUBLIC AEROSPACE– *Java Integration Developer, Kennesaw, Georgia, USA*

February 2024 – Present

- Configured and managed Azure APIM endpoints to receive and process data from a third-party application, increasing the efficiency of handling incoming data by 30%, reducing latency and improving data throughput.
- Engineered and optimized workflows using Azure Logic Apps and automated data processing with Azure Function Apps to route and handle data across various TPA's and downstream systems, reducing manual intervention and processing times by 75%.
- Streamlined Azure Service Bus for reliable messaging, enhancing integration with downstream systems and improving delivery reliability by 35%, ensuring timely and accurate data distribution

VYSHNAVI INFORMATION TECHNOLOGIES – *Software Engineer, Bangalore, India*

August 2019 – July 2021

- Optimized the Dual Intent and Entity Transformer classifier by 20% using RASA to classify intents and entities like name, locations, and phone numbers
- Designed two chatbots, one was a restaurant bot which served to handle customer orders and billing
- Implemented a taxi bot which helped customers book a cab, and get their ride histories
- Devised React components for the RPA tool Intellibuddies to cut development costs by 30%

## ACADEMIC PROJECTS

SEAM CARVING FOR IMAGE RESIZING AND OBJECT REMOVAL

Spring 2022

Project Link: <https://github.com/Himanshu190397/Seam-Carving-And-Object-Removal-From-Scene>

- Dynamically inserted and removed low energy seams from an image using pytorch to resize it using forward and backward energy which preserved the aspect ratio, image content and image quality compared to traditional image resizing techniques
- Synthesized a binary mask to remove an object from the scene

## STROKE PREDICTION USING VANILLA MACHINE LEARNING MODELS

Spring 2022

Project Link: [https://github.com/Himanshu190397/Stroke\\_Prediction\\_](https://github.com/Himanshu190397/Stroke_Prediction_)

- Created Machine Learning Classifiers without importing pre-built libraries that can predict whether a person could be diagnosed with a stroke or not
- Modelled Logistic Regression, K-means clustering, Naïve Baiyes and Decision Tree and used precision as the metric to compare them

## BREAST CANCER PREDICTION

Winter 2022

Project Link: <https://github.com/Himanshu190397/Breast-Cancer-DetectionWisconsin>

- Developed a tool to predict cancerous masses based on various parameters using Breast Wisconsin Cancer dataset.
- Juxtaposed Random Trees, Gradient Boosted Trees, KNN and Logistic Regression, to identify that Logistic Regression outperformed the rest with a Recall value of 98%

## COVID IMPACT ON MENTAL HEALTH

Fall 2021

- Devised a data acquisition and pre-processing pipeline, which pulled 1 million tweets per hour relating to Covid and Mental Health using snsrape
- Pre-processed these tweets and pushed it into a MongoDB collection running on the cloud which can be used by hospitals and academia for further analysis and research